Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/01/2006
PI Name:	Maxworthy, Tony Ph.D.		
Project Title:	The Dynamics of Miscible Interfaces: A Space Flight Experiment		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	90089-1453	**Congressional District:**	33
Comments:	9/7/16 - PI is deceased (March 8, 2013) per online university obituary.		
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-01
Start Date:	01/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Balasubramaniam, R.	**Contact Phone:**	216-433-2878
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2006 (from 12/31/2006) per PI information in 2007 (1/2009) Note: grant renewed 4 yrs		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04GB29G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Both theoretical and experimental research will be performed. In the experimental studies, a miscible liquid displaces another liquid in a cylindrical tube. The displacement velocity is such that the Peclet number for the mass transfer is large compared to unity. The velocity and concentration fields near the mixing front are obtained by particle image velocimetry and interferometry respectively. The experimental results will be compared in-depth with computational results that include Korteweg stresses and the non-vanishing divergence of the velocity field in the mixing region. The computations will predict the spatio-temporal evolution of the velocity and concentration fields for a variety of system parameters. The objective of this research is to try to unravel a number of difficult issues that are present in the study of miscible interfaces. It is known that in addition to the usual stresses in a fluid, additional stresses called Korteweg stresses could be important in regions where the concentration gradients are large. Also, in the mixing region, the velocity field is not divergence free in incompressible fluids. The goal of this research is obtain the flow and concentration fields near miscible interfaces in microgravity and compare them with computational results that include the Korteweg stresses and divergence. The improved understanding of the dynamics of mixing that will be obtained from this research can be applied to multiphase flows in porous media such as those occurring in enhanced oil recovery, fixed bed regeneration, hydrology and filtration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The task has been cancelled due to the NASA reorientation to the Moon/Mars Project. Funding was continued to allow us to complete the ground based lab. work. A brief summary of this is given below: This report presents experiments of a less viscous liquid displaced by a more viscous, miscible liquid in a vertical tube of small diameter. The liquids used are silicone oils of various viscosities. The more viscous fluid has a higher density than the less viscous fluid. The experiments showed that the interface between the liquids can be unstable and exhibits an asymmetric, sinuous shape for downward displacement. The Reynolds number is negligibly small in these experiments and the Peclet number for mass transfer is O(50)–O(105). Visualization and PIV measurement of the unstable finger are reported. Experiments were also conducted to study immiscible displacements in tubes in order to compare with previously obtained results for miscible fluids. Three immiscible fluid pairs, 50cst silicone oil/96% glycerine (At=0.81), 500cst silicone oil/H-15 polybutene (At=0.44) and 500cst silicone oil/H-25 polybutene (At=0.81), were used to obtain a range of capillary numbers from 0.01 to about 5. The experiments were conducted in 1mm and 2mm inner diameter horizontal tubes. The experiment showed no finger in the tube at a very low Capillary number, i.e., Ca<0.01, here the Capillary number defined as Ca=Umμ2/s, Um is the mean velocity in the tube, μ2 the viscosity of displaced fluid and s the interfacial tension. The fraction m, obtained by measuring the finger propagating speed, increased rapidly at small Ca and then tends to a constant value at large Ca, depending on At. The measurement shows the value for the fluid pair 500cst silicone oil/H-15 polybutene is 0.5 and for 500cst silicone oil/H-25 polybutene is 0.55, these values are consistence with the result for miscible case at large Peclet number. The fraction m was also obtained by measuring the thickness of displaced fluid left on the tube wall behind the finger tip. This value was found to be larger than that using the finger speed. Possible reasons for this include: the finger may not be exactly symmetric in the tube due to the effect of gravity, even though small,, the fact that the finger may not have a constant diameter along its length and the fact that surface tension acts to modify the finger tip. DPIV was used to measure the velocity flow field around the finger tip for immiscible displacements in a round tube. An upward displacement of a less viscous fluid into a more viscous fluid in a vertical tube was conducted in a 7mm diameter tube of 15cm long. Two fluid pairs: 500cst silicone oil/H-25 polybutene (At=0.81) and 10M silicone oil/H-25 polybutene (At=0.38) were used. For 500cst silicone oil/H-25 polybutene, gravity distorted the finger as the displacing fluid had a larger density and a fat finger was found with m=0.23 (injected flow rate Q=10ml/h, Ca=0.014 and F˜200). The velocity field showed a recirculation region in the finger tip. This result agrees with our previous conclusion for miscible fluid for m<0.5. In the cancelled MIDAS space-borne experiment, low viscous fluid (LVF) was to have been injected from the inlet end of a tube filled with high viscous fluid (HVF) so as to generate a finger and, then, to flush the tube using the same HVF for next experiment. Previous visualization showed that a tail behind the flushed LVF bubble remained in the HVF to contaminate it. An experiment was conducted to study the effect of contamination on the finger flow field in a round tube by measuring the velocity field using DPIV. A special cap with two nozzles was made for the inlet end of the round tube. One nozzle was to inject LVF (displacing fluid) for a fingering displacement and the other shorter nozzle was to inject the flushing fluid (the same HVF). Fluid at the outlet of the round tube was drained into a beaker. The tube was mounted vertically and upward displacements were performed. After measuring the velocity field with a displacing finger in a clear HVF, the tube was flushed using the same high viscous fluid. Then, another measurement was conducted at the same location to investigate any difference between the velocity fields before and after flushing. The results showed that the contamination presented no effect on the finger velocity field if the flushing was complete. However, if the flushing was not complete, e.g., if there was a small amount of LVF remaining around the nozzle orifice, the displacing fluid would penetrate through a narrow channel and move at a very high velocity. Although the finger would eventually be generated if the tube was long enough, the flow field might be much different from the one for clear HVF. In a ground experiment this problem could be worse as the light displacing fluid would also have a buoyancy induced velocity. In a space experiment, with no buoyancy, the contamination of HVF due to flushing will not affect the velocity field if the flushing is complete. We should emphasize here that the present experiment was carried out at a very high displacement speed where the diffusion had less effect on the flow field. If displacement is at a very slow speed and diffusion dominates the dynamics in the flow, we are still not sure if the contamination of LVF in the HVF has any effect on the finger flow field. Our experimental observations show that in the case of miscible displacement of a less viscous fluid by a more viscous one, several interesting regimes of flow and shapes of the interface demarcating the fluids occur. A quantity of chief interest is the efficiency of the displacement process measured, for example, by the fraction of the displaced fluid left behind on the tube wall. In this report, we present experimental results of the displacement of a less viscous liquid by a more viscous, miscible liquid in a vertical tube. The motivation for our work arose from the design of a space experiment to study the dynamics of miscible fluids, with a range of viscosity ratios. The displacement of a more viscous liquid by a less viscous one and the reverse displacement of a less viscous liquid by a more viscous one was

	encountered in the design of the space experiment. Interfacial stability between two miscible liquid is an interesting subject to study in the space project. One important issue regarding this stability is the unstable finger we observed previously for immiscible displacement in capillary tubes. This instability is thought to be a capillary instability due to the weak interfacial tension between two fluids. In this experiment, a syringe was used to displace the fluids. An Oil-Blue dye was added to the more viscous liquid for ease of observation of the interface between the fluids. The dye did not have a measurable effect on the density of the liquid. The tube was first filled to the desired level by pumping the desired fluid (less viscous fluid for downward displacement; more viscous fluid for upward displacement) from the bottom of the tube. A small initial distortion of the interface was unavoidable. By rotating the tube while obtaining the images as well as observing the flow from two orthogonal views, we have determined that the asymmetry is two dimensional, and not helical in nature. Selected experiments were performed in tubes of larger diameter of 7mm and 10mm to measure the velocity fields using particle image velocimetry (PIV). A detail description of this experimental setup is provided in Kuang, Petitjeans, and Maxworthy. In this experiment, a more viscous fluid of 1000 cSt silicone oil displacing 100 cSt silicone oil in a 10 mm diameter tube displacing a more viscous was conducted, a sinuous shape of the finger body was clearly observed.
Bibliography Type:	Description: (Last Updated: 01/26/2009)
Articles in Peer-reviewed Journals	Balasubramaniam R, Rashidnia NT, Maxworthy T, Kuang J. "Instability of miscible interfaces in a cylindrical tube." Physics of Fluids 2005 May;17(5):052103. http://dx.doi.org/10.1063/1.1884645 , May-2005
Articles in Peer-reviewed Journals	Kuang J, Maxworthy T, Petitjeans P. "Miscible displacement between silicone oils in capillary tubes." European Journal of Mechanics - B/Fluids, 2003 May-Jun;22(3):271-7. http://dx.doi.org/10.1016/S0997-7546(03)00035-9 , Jan-2004
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